

SHIVAM INDORE

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RESEARCH SUMMARY

B.Tech Computer Science student (6th semester, FAST–JLU Bhopal) with 1.5+ years of hands-on ML/AI research experience. Built and deployed a pharmacogenomic drug recommendation system (Genovix) on 300K+ patient records using XGBoost with ~80% validated accuracy and extended it with an LLM reasoning layer (Groq/Llama 3). Experienced in tabular ML pipelines, FastAPI backend development, and integrating large language models for clinical NLP. Seeking a research internship to contribute to frontier AI problems in explainability, tabular foundational models, and alignment.

EDUCATION

B.Tech — Computer Science and Engineering

Aug 2023 – Expected May 2027

FAST – JLU (Jabalpur Lakshmikant University), Bhopal, India | Roll No. 2023BTCSE016

- Relevant coursework: Machine Learning, Data Analysis & Visualization, Theoretical CS, Algorithms, Database Systems
- Active participant in national-level hackathons under team HACKFINITY

TECHNICAL SKILLS

ML / AI Frameworks: Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, TensorFlow (basic)

LLM & NLP: Groq API, Llama 3, prompt engineering, chain-of-thought reasoning, conversational agent design

Backend & APIs: FastAPI, Django, REST APIs, JWT authentication

Frontend: Next.js, React, Vite.js, TypeScript, Tailwind CSS

Tools & Practices: Python, Git, Jupyter Notebook, Power BI, Docker (basic), CI/CD basics, modular code design

Research Areas: Tabular ML, pharmacogenomics, model interpretability, uncertainty estimation, LLM alignment

RESEARCH PROJECTS

Genovix — AI-Based Pharmacogenomic Drug Recommendation System

May 2025 – Present

Minor Project, FAST–JLU Bhopal | Stack: XGBoost · FastAPI · Next.js · Groq API · Llama 3

- Designed and trained a drug recommendation pipeline on 300,000+ patient records spanning 15 drug classes using XGBoost, achieving ~80% validated accuracy with stratified cross-validation
- Engineered a 49-feature input schema with MinMaxScaler, LabelEncoders, and pickle-serialized model artifacts for reproducible inference via a FastAPI REST API
- Identified and eliminated critical data leakage via the Treatment_Type column; rebuilt the evaluation pipeline with stratified k-fold CV, reducing inflated accuracy from ~95% to a genuine ~80%
- Extended the system with a Groq/Llama 3 LLM layer for natural-language patient intake, clinical reasoning generation, and conversational follow-up — transforming it from a pure ML classifier into an AI diagnostic assistant
- Presented at university expo; prepared SIH-style submission and national hackathon pitch deck (Oriental TechHack 2.0, ₹1,00,000 prize pool)

Indian House Price Prediction — ML Research Pipeline

Jan 2026

ML Course Project | Stack: Python · Scikit-learn · XGBoost · Jupyter Notebook

- Built end-to-end supervised learning pipeline on India Housing Prices Dataset: EDA, log transformation, IQR-based outlier removal, LabelEncoder, StandardScaler, and GridSearchCV hyperparameter tuning
- Benchmarked Random Forest, XGBoost, and Gradient Boosting; incorporated K-Means clustering as an unsupervised feature to improve regional price segmentation
- Debugged LabelEncoder re-use bug (single le object overwritten in loop), resulting in correct category mappings and improved model generalization

VoiceNav — Offline Voice Navigation Assistant

Mar 2026

KRIYETA 5.0 Hackathon · Team HACKFINITY | Stack: Whisper.cpp · OSMnx · A · pyttsx3*

- Built a fully offline-first voice assistant for in-car navigation using Whisper.cpp for STT, regex-based NLP intent parsing, and A* pathfinding over OpenStreetMap graphs — zero internet dependency
- Designed modular architecture enabling sub-second intent resolution and turn-by-turn guidance with pyttsx3 TTS

AgriSense AI — Precision Farming Assistant

Apr 2026

Oriental TechHack 2.0 · Team HACKFINITY | Stack: FastAPI · TensorFlow · LangChain · Groq

- Architected a two-module AI platform: a CNN/ViT plant disease detector and a Groq + Llama 3 RAG advisory agent for crop management recommendations
- Designed end-to-end system from data ingestion to farmer-facing conversational interface, targeting rural deployment with low-latency inference

HACKATHONS & COMPETITIONS

- HackWithAI — Top 20 Teams, Telangana's Largest 24-Hour AI Offline Hackathon (KLH University, Hyderabad) — Certificate of Merit

ADDITIONAL EXPERIENCE

- Power BI Dashboards: Built IBM HR Analytics attrition dashboard and World University Rankings dashboard with DAX measures, KPI cards, drill-through, forecasting, and data cleaning in Power Query
- FIFA 22 Multivariate Analysis: Comprehensive statistical analysis on 19,239 records × 110 features; identified Overall Rating–Market Value correlation of $r = 0.852$ using heatmaps, violin plots, and parallel coordinates
- EventFlex: Django 5.2 event management platform with REST API, JWT auth, and QR code check-in system

RESEARCH INTERESTS

Tabular Foundational Models · Explainability & Trust (XAI) · LLM Alignment · Pharmacogenomics AI · Uncertainty Estimation · Post-training Fine-tuning (LoRA / Instruction Tuning)